PACLAND · Climate

The verification discipline applied to climate modelling — a plain-English brief

The headlines — and what to do about each

- 2 models leak heat at ~1.9× the signal — don't build on the leakers, use the 5 that conserve all three laws.
- A 2100 projection is extrapolation, not verification (0 future states measured) — spend on measurement, not compute.
- The leak is in the coupling, not the ocean (12/19 energy, 6/19 mass, 2/19 salt) — fix the air-sea exchange, the ocean model is fine.
- Only 2 of 19 models survive all three axes — conservation-clean, reasonable sensitivity, reasonable forcing (GISS-E2-1-G-CC, GISS-E2-1-G).
- The forcing axis sharpens it: MPI-ESM1-2-LR / HAM leak AND force too strongly, ACCESS-ESM1-5 / BCC / CNRM-ESM2-1 leak AND force too weakly — conservation alone is never enough.
- Skillful ≠ true — ACE's '>90% of tracked variables' is 93% model-vs-model (41/44 vs a coarser GCM); against the Earth it is certified only to ~96% on m=2 observables.
- The best-prediction ECS: 4.08 K naive → 3.86 K conservation-gated → 2.71 K trust-gated — the verification gates remove 1.37 K of hot-model bias and land at the IPCC best estimate.
- Trust by verdict, not by reputation — a machine-checked pass/fail, not a paper footnote.

1 · What we are measuring

Climate projections are the most consequential predictions we make. The discipline asks the question the field has never answered in numbers: how much of a projection is verified, versus extrapolated?

2 · The verification deficit

- 4 of 12 historical observables are measured + computable + validated — a certificate radius of 68%.
- 0 of 5 2100-projection targets are validated — the projection is extrapolation, not verification.

A 1% certificate needs ~18,000 validated observables — three orders of magnitude beyond what the historical record can supply. The bottleneck is measured data, not compute.

3 · The energy budget, on real data

- The 0–2000 m ocean alone under-closes the budget (residual 0.15 W/m²).
- The full budget — ocean + atmosphere + land + cryosphere — closes it (residual 0.01 W/m²). An

invariant is only an invariant when all its terms are counted.

4 · The CMIP6 conservation drift — three ledgers

A conserving model must close three independent ledgers in its control run: energy, mass, and salt. The measured drift shows how many fail each.

- Energy: 12 of 19 models leak (net TOA beyond 0.1 W/m²); 2 leak faster than the EEI — CNRM-ESM2-1 at 1.485 W/m² (1.93×).
- Mass: 6 models leak ocean mass — ACCESS-ESM1-5 at -1.03 mm/yr, 30% of observed sea-level rise.
- Salt: only 2 models leak — salt is conserved orders of magnitude better.
- So the leak lives in the atmosphere-ocean coupling, not the ocean model: 5 of 19 models conserve all three ledgers.

5 · Why it matters

A model leaking heat at ~2× the signal it is projecting carries an error the size of the projection. Conservation is not a solved detail — it is a hard, quantified defect in most models, invisible to skill scores and to ensemble spread. And because salt (purely ocean) is conserved far better than energy and mass (which cross the air-sea boundary), the defect is in the coupling, not the ocean.

6 · Reproduce

pacland climate · python3 scripts/climate_cmip6_drift.py · docs/climate-cmip6-drift.md